Cross-Scene Hyperspectral Image Incremental Classification Based on Textual Knowledge Guidance and Dual Distillation

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***Abstract*—Class-incremental learning (CIL) is of significant value for dynamic Earth observation applications using hyperspectral images (HSIs). However, existing methods addressing catastrophic forgetting often fail to simultaneously handle limited storage budgets, interclass semantic ambiguity, and the effective balance between new and old knowledge. To systematically address these challenges, this article proposes a novel CIL framework based on text guidance and dual distillation (TGDD). First, a distribution-aware sample selection strategy is designed to address the conflict between the inherent intraclass spectral variability of HSI data and limited storage budgets. This strategy adaptively discovers the intrinsic substructures of each class and ensures that they are effectively represented in the exemplar set, thereby maximizing the fidelity of historical knowledge under strict storage constraints. Second, a dual distillation mechanism is constructed to achieve a balance between knowledge "retention" and "learning." Specifically, intertask knowledge distillation (InterKD) transfers the discriminative ability of the old model to the current one without disrupting consolidated knowledge. Meanwhile, intratask hierarchical self-distillation (IntraHD) promotes collaborative learning between deep and shallow layers, ensuring consistent and highly discriminative feature representations. Furthermore, to tackle misclassification caused by spectrally similar classes across new and old tasks, a multimodal semantic guidance (MSG) module is introduced. Based on text embeddings of class names, this module dynamically adjusts feature learning and decision boundaries, enhancing the model's ability to distinguish between spectrally similar but semantically different classes. Extensive experiments on four public HSI datasets demonstrate that TGDD achieves a superior balance between stability (old knowledge) and plasticity (new knowledge). Its classification accuracy and robustness significantly outperform current state-of-the-art methods. The code is available at https://github.com/Li-ZK/TGDD**

***Index Terms*—Dual distillation, hyperspectral image (HSI), incremental learning, multimodal learning.**

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NOMENCLATURE

CASI	Compact airborne spectrographic imager.
CIL	Class-incremental learning.
CRD	Classification reverse distillation.
DAES	Distribution-aware exemplar selection.
FRD	Feature reverse distillation.
InterKD	Intertask knowledge distillation.
IntraHD	Intratask hierarchical self-distillation.
MSG	Multimodal semantic guidance.
ROSIS	Reflective optics system imaging spectrometer.
TGDD	Text guidance and dual distillation.

I. INTRODUCTION

HYPERSPECTRAL images (HSIs), characterized by hundreds of continuous spectral bands, provide rich spectral information that makes them invaluable in fields, such as land cover mapping, ecological monitoring, and mineral exploration [1], [2], [3]. However, the inherent high dimensionality and spectral complexity of HSIs pose significant challenges for accurate classification. Recently, deep learning (DL), with its powerful nonlinear feature extraction capabilities, has become a cutting-edge technique to break through the bottlenecks of traditional classification methods. By automatically learning deep coupled spectral–spatial features, DL models have significantly improved HSI classification accuracy [4], [5], [6], and a variety of deep architectures have since been developed to further enhance the robustness and discriminability of such spectral–spatial representations [7], [8], [9].

With the rapid development of remote sensing technology, spaceborne sensors continuously acquire massive, dynamic Earth observation data on a global scale. This trend requires classification models to not only handle static datasets but also possess human-like continual learning capabilities—learning knowledge of new object types while robustly retaining historical knowledge. However, traditional DL models generally

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suffer from "catastrophic forgetting" in sequential tasks, where the performance on old tasks degrades sharply after learning new ones. To address this challenge, CIL has emerged [10], [11], [12], [13]. The core of CIL lies in seeking an effective balance between model *plasticity* (the ability to adapt to new classes) and *stability* (the ability to prevent forgetting of old knowledge). This paradigm is particularly critical for HSI classification tasks that involve monitoring the dynamic evolution of ground objects.

To mitigate catastrophic forgetting in CIL, researchers have proposed various strategies. These methods can be broadly categorized into four types: 1) replay-based methods, which store a small number of representative historical samples in a limited memory budget and replay them during new task training to maintain the data distribution of old tasks [14], [15], [16], [17]; 2) regularization-based methods, which impose penalty terms on changes in key model parameters to restrict drastic shifts during new task learning, thereby protecting learned knowledge [18], [19], [20]; 3) knowledge distillation (KD)-based methods, which transfer knowledge (e.g., output probabilities or intermediate features) from an old model (teacher) to the new model (student) to achieve intergenerational knowledge transmission [21], [22], [23], [24]; and 4) bias correction-based methods, which focus on adjusting the output layer to correct the natural bias toward new classes, ensuring fair prediction between new and old categories [25].

Although the aforementioned strategies have achieved some success in general computer vision and have been preliminarily applied to HSI incremental classification [26], [27], [28], they still face several severe and interconnected challenges in complex remote sensing scenarios.

First, there is an inherent conflict between intraclass spectral variability and storage budgets. In long-term on-orbit missions, the storage space allocated to each historical class is continuously compressed as incremental steps accumulate. Simple random sampling (RS) or fixed-quantity replay strategies fail to capture complex intraclass variability (e.g., distinct spectral characteristics of the same object class caused by varying imaging conditions, object states, and spatial heterogeneity). This leads to severely insufficient representation in the exemplar set, thereby accelerating catastrophic forgetting [29], [30]. Moreover, existing methods often overlook the inherent substructures of hyperspectral data. Treating each class as a single distribution for sampling results in the loss of key patterns when dealing with remote sensing data characterized by significant intraclass diversity.

Second, semantic ambiguity between hyperspectral classes exacerbates knowledge confusion. HSIs often contain spectrally similar "fine-grained" classes (e.g., "forest" versus "shrubs"). When a new class highly overlaps spectrally with an old class, the model is prone to interfering with or even overwriting the feature representation of the old class during learning, leading to knowledge confusion rather than knowledge accumulation [31]. Existing methods primarily impose constraints within the visual feature space. They lack guidance from the semantic level, making it difficult to fundamentally resolve the issue of interclass spectral confusion.

Finally, the effectiveness of knowledge transfer is often insufficiently coupled with the model architecture. Designing a network architecture that effectively consolidates old knowledge, while efficiently learning new knowledge, and balancing the relationship between the two, remains an open problem. On the one hand, direct KD may cause gradient conflicts on the feature extractor between new and old tasks, failing to thoroughly protect old knowledge. On the other hand, inconsistent feature representations across different layers may occur during new task learning, hindering the generalization capability for new knowledge.

To systematically address these challenges, this article proposes a novel CIL framework based on TGDD. By integrating sample selection, KD, and multimodal learning, this method aims to achieve more robust HSI incremental classification performance. The main contributions of this study are summarized as follows.

1) A distribution-aware sample selection strategy is proposed. Unlike conventional RS or centroid-based selection methods that treat each class as a single compact distribution, the proposed strategy adaptively discovers the intrinsic substructures within each class through unsupervised clustering. This ensures that the complex distributional characteristics of the original data are preserved under strict memory constraints, thereby maximizing the representational fidelity of the exemplar set with respect to historical knowledge.
2) A dual KD module is designed, which transfers the discriminative capability of the old model to the current model via InterKD without disrupting the learning of new knowledge. Simultaneously, intratask self-distillation promotes collaborative learning between the shallow and deep layers of the network, ensuring the consistency and high discriminability of feature representations. The synergistic operation of these two mechanisms enables the model to effectively retain old knowledge while acquiring new knowledge.
3) An MSG module is introduced, which leverages text embeddings of category names as semantic priors to assist the model in distinguishing similar classes. At each incremental learning stage, this module dynamically adjusts feature learning and decision boundaries. By incorporating semantic knowledge, it substantially alleviates the confusion encountered by purely visual feature methods when differentiating classes that are spectrally similar but semantically distinct, thereby enhancing class discrimination capability in incremental learning scenarios.

The remainder of this article is organized as follows. Section II introduces the related work. Section III details the proposed TGDD method. Section IV presents the experimental results on four public HSI datasets. Finally, Section V concludes this article.

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II. Related Work

A. Sample Selection Strategies

Replay-based methods mitigate catastrophic forgetting by storing a small set of representative samples from historical tasks in a memory buffer and replaying them during new task training so that the data distribution of old tasks is preserved.

Regarding sample selection mechanisms, Lopez-Paz and Ranzato [32] proposed the gradient episodic memory (GEM) method. This approach quantifies the contribution of samples to model gradients, prioritizing the retention of key samples that minimize the forgetting of historical tasks. Rebuffi et al. [14] introduced the incremental classifier and representation learning (iCaRL) algorithm. It utilizes a cosine similarity metric to select samples that are closest to the class mean vector and most representative of the category to construct the exemplar set. Furthermore, Isele and Cosgun [33] designed a sample importance evaluation criterion based on information entropy theory. This method ranks samples according to their potential contribution to current and future tasks, prioritizing the replay of samples with high entropy values during training. However, HSIs exhibit complex intraclass substructure characteristics due to factors such as varying imaging conditions, differences in land cover states, and spatial heterogeneity. This makes it difficult for sample selection strategies based on a single structure to adequately represent their complex distribution characteristics.

Masana et al. [34] systematically compared random, class-boundary, and herding-based selection and showed that under imbalanced distributions, none of them reliably preserves the statistical properties of the original data distribution with a limited number of memory samples. To address these limitations, the distribution-aware sample selection strategy proposed in this article adaptively discovers the inherent substructures of each class through unsupervised clustering and selects the most representative samples from each substructure. This method not only maximizes sample diversity under limited storage but also fully represents hyperspectral data with complex distribution characteristics.

B. KD for Incremental Learning

KD [35], [36] as a classic knowledge transfer technique, effectively suppresses knowledge forgetting during new task learning by transferring knowledge from a pretrained teacher model to a student model. According to the level of knowledge transfer, KD methods can be categorized into three types [37]: response distillation, feature distillation, and relation distillation.

Li and Hoiem [38] pioneered the introduction of KD into CIL with the learning without forgetting (LwF) algorithm. Building on LwF, iCaRL [14] incorporated a sample replay mechanism, enforcing prediction consistency between the new and old models on historical classes to strengthen knowledge retention. Considering the imbalance in the number of new and old classes across different incremental stages, BiC [25] introduced a bias correction layer to prevent the model from developing an excessive preference for new tasks. Additionally, several studies have successfully transferred the LwF framework to HSI classification tasks [23], [26], [28].

In the direction of feature distillation, AFC [39] performs adaptive distillation based on the importance of different feature maps. PODNet [40] achieves knowledge transfer by minimizing the spatial discrepancies of intermediate layer features after pooling. GeoDL [41] constructs distillation objectives based on feature projection geometric relationships. While response and feature distillation primarily address instance-level knowledge mapping, researchers have proposed relational KD to further mine structured knowledge within models [42]. For instance, ERL [43] extends relational regularization to few-shot CIL scenarios. TPCIL [44] constraints knowledge evolution by modeling the topological relationships of elastic Hebbian graphs. MBP [45] enforces consistency in distance relationships between the new and old models at the instance neighborhood level. Although these KD methods have achieved success in general computer vision, they typically operate at a single knowledge level and lack a task-adaptive mechanism to balance old knowledge retention against new knowledge learning, which limits their effectiveness in hyperspectral incremental classification. To this end, this article proposes a dual KD module. Through knowledge transfer units aligning the knowledge of the old and new models at different abstraction levels and an intratask self-distillation mechanism promoting the collaborative learning of shallow and deep features, this module enables the model to effectively retain old knowledge while fully acquiring new knowledge, achieving a dynamic balance between retention and learning.

C. Multimodal-Guided Classification

In recent years, vision-language pretrained models have demonstrated immense potential in the field of continual learning. By performing joint pretraining on large-scale image–text pairs, these models learn cross-modal semantic alignment representations and have achieved significant performance in various downstream vision and language tasks.

Based on architectural design, mainstream vision-language models can be divided into single- and dual-stream architectures. Single-stream architectures concatenate image and text embeddings and model them jointly through a unified Transformer network to build a shared multimodal semantic space. Representative works include VisualBERT [46] and unified modal pretraining models [47]. Dual-stream architectures employ independent image and text encoders to extract high-level semantic features. Typical representatives include the contrastive language-image pretraining (CLIP) model [48] and CoCa [49]. These models adopt contrastive learning as the training objective. By constructing positive and negative sample pairs, they bring semantically related cross-modal samples closer in the representation space while pushing unrelated samples apart, thereby enhancing the model's cross-modal representation capability.

However, in continual learning scenarios, since all information is processed within a shared space, the training of new tasks can easily interfere with the representations of historical tasks, leading to catastrophic forgetting. To alleviate this issue, Qiu et al. [50] proposed combining vision-language foundation

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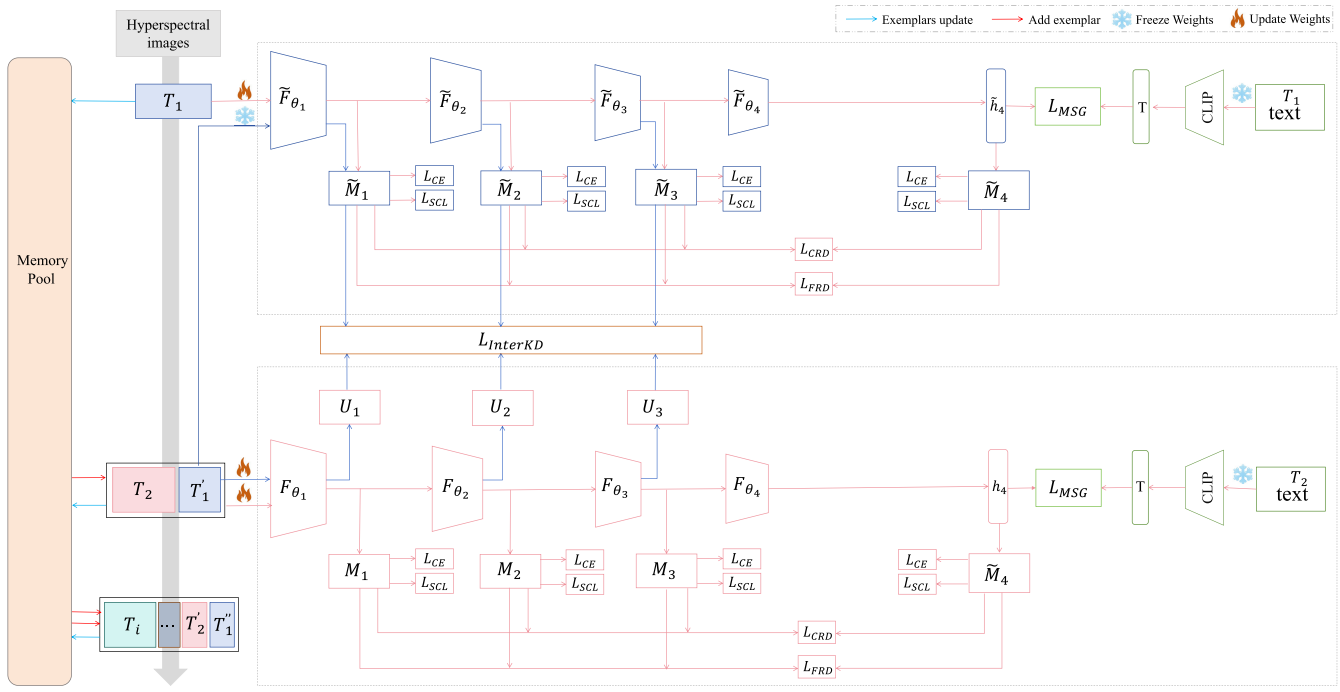


Fig. 1. Framework of the proposed TGDD method. The training process consists of two phases: the base task phase and the incremental task phase. First, in the base phase, the model is trained on initial classes using the IntraHD and MSG modules to learn robust feature representations. Subsequently, in the incremental phase, the model integrates DAES, InterKD, IntraHD, and MSG modules. This combination enables the effective learning of new tasks while consolidating the knowledge representation of old tasks, thereby mitigating catastrophic forgetting and class confusion.

models with memory replay strategies, storing a small number of historical samples or their encoded representations in a buffer for knowledge review during new task training. This method enables the model to effectively maintain the representation quality of historical tasks while learning new ones, significantly reducing the degree of knowledge forgetting. However, existing methods relegate the text modality to a passive auxiliary role and fail to fully exploit its discriminative semantic information. In HSI classification, interclass confusion caused by the spectral similarity of land covers is particularly prominent. Static text replay strategies struggle to dynamically utilize cross-modal semantic constraints to enhance the separability of hard-to-classify samples during the model learning process, causing the class boundaries between new and old tasks to gradually blur in incremental learning. This article proposes an MSG module. Through explicit text–image feature alignment, it ensures that spectrally similar but semantically distinct classes maintain sufficient discriminative distance in the representation space, effectively mitigating catastrophic forgetting and interclass confusion in hyperspectral incremental learning.

III. Proposed Method

To address the prevalent issue of "catastrophic forgetting" in HSI CIL and resolve class confusion caused by spectral similarity, this article proposes a novel TGDD. The proposed framework integrates four core components designed to efficiently acquire new knowledge while robustly retaining the memory of historical knowledge within a dynamic environment where new tasks are continuously introduced. Specifically, these components are: 1) DAES, which aims to maximize the fidelity of historical knowledge under a limited storage budget; 2) InterKD, which utilizes a knowledge transfer unit to efficiently transfer knowledge from the old model; 3) IntraHD, which enforces the model to learn hierarchically consistent and highly discriminative features; and 4) MSG, which leverages textual priors to mitigate class confusion. The overall framework is shown in Fig. 1.

A. DAES Module

Traditional exemplar selection methods, such as RS or the Herding algorithm, typically assume that each class follows a single compact distribution. However, in HSI scenarios, due to factors such as illumination variations, distinct growth states of ground objects, and spatial heterogeneity, the same class often exhibits significant *intraclass substructures*. This means that its features form multiple separated clusters in the latent space. Ignoring this complex distribution characteristic results in the exemplar set failing to comprehensively cover the decision boundaries of the class, thereby exacerbating the forgetting of old knowledge during the incremental learning process.

To this end, we propose a DAES strategy. The core idea is to automatically discover the intrinsic substructures of each class via unsupervised clustering and select the most representative samples from each substructure. This ensures that the exemplar set faithfully reflects the complex distribution of the original data under a strict storage budget.

Assume that after the tth incremental stage, the model has learned C_t classes, with a total storage budget of M samples.

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For any class $c \in \{1, \ldots, C_t\}$, let the feature set of all its training samples be $f_c = \{\widetilde{F}_\theta(x) \mid x \in \mathcal{D}_c\}$, where $\widetilde{F}_\theta(\cdot)$ is the frozen feature extractor of the old model and $\mathcal{D}_c$ is the complete training set for class c. The DAES strategy executes the following steps.

1) Adaptive Substructure Discovery: First, we assign a sample quota $m_c = \lfloor M/C_t \rfloor$ to each class c. Then, we apply the K-means clustering algorithm to the feature set f_c, partitioning it into m_c subclusters $\{S_c^1, S_c^2, \ldots, S_c^{m_c}\}$. Each subcluster S_c^k represents a potential intraclass pattern, and its centroid is calculated as

$$\mu_c^k = \frac{1}{|S_c^k|} \sum_{x \in S_c^k} \widetilde{F}_\theta(x). \tag{1}$$

By binding the number of clusters to the sample quota, DAES adaptively allocates representation resources for each class.

2) Substructure Representative Sample Extraction: For each discovered subcluster S_c^k, instead of selecting the centroid itself (which is a mean vector and may not correspond to a real sample), we select the original sample in the feature space that is closest to the centroid μ_c^k as the representative of that substructure

$$x_c^k = *argmin_{x \in S_c^k} \left\| \widetilde{F}_\theta(x) - \mu_c^k \right\|_2. \tag{2}$$

This method ensures that each selected sample is located in the core region of its corresponding data pattern, thereby maximizing the information entropy and representativeness of individual samples.

3) Global Exemplar Set Construction: Repeating the above process for all historical classes, the final constructed exemplar set $\mathcal{M}^t$ is the union of representative samples from all classes

$$\mathcal{M}^t = \bigcup_{c=1}^{C^t} \left\{x_c^1, x_c^2, \ldots, x_c^{m_c}\right\}. \tag{3}$$

This set satisfies the storage constraint $|\mathcal{M}_t| \le M$ and maximizes the coverage of the complex distributions of historical classes under the limited budget. Unlike the iCaRL Herding algorithm, which only focuses on class centers, DAES not only preserves global class information but, more importantly, maintains the local structure of intraclass distributions. This provides a higher quality knowledge source for subsequent distillation learning, laying a solid foundation for accurately reconstructing the decision boundaries of old classes.

B. Dual Distillation Module

KD is an effective means of mitigating catastrophic forgetting in CIL by transferring knowledge from the old model to the new model. However, in hyperspectral incremental classification, existing distillation strategies still face the following core challenges. First, most methods impose constraints at only a single knowledge level: response-level distillation focuses on the model's output decisions, whereas feature-level distillation emphasizes the representation consistency of intermediate layers. The two lack organic coupling, making it difficult to simultaneously ensure the coordinated optimization of discriminative prediction and semantic representation. Second, existing methods fail to effectively balance the retention of old knowledge with the learning of new knowledge: excessively strong distillation constraints suppress the model's ability to learn new tasks, while excessively weak ones are insufficient to resist catastrophic forgetting, lacking a task-adaptive adjustment mechanism. Furthermore, existing methods primarily focus on cross-task alignment between the old and new models, neglecting the complementary relationship between deep abstract features and shallow general features within the same task, which results in a lack of consistency in the hierarchical representation of new classes.

To this end, we design a dual KD mechanism to synergistically alleviate the aforementioned issues from both the intertask and intratask dimensions. InterKD aligns the knowledge of the old and new models at different abstraction levels to transfer old knowledge to the new task, while IntraHD performs feature-level and classification-level reverse distillation within a single incremental stage to promote the collaborative learning of shallow and deep features. This enables the model to fully acquire new knowledge while effectively retaining old knowledge. Both InterKD and IntraHD are built upon a universal adapter module.

1) Adapter Module: The adapter serves as a bridge connecting the features from different levels of the backbone network to the downstream tasks. For the feature map h_i output by the ith layer of the current task's feature extractor F_θ, we design a corresponding adapter M_i. As shown in Fig. 2, the adapter M_i comprises the following.

1) *Dimension Alignment Layer W_i:* A linear layer that maps the feature h_i from the ith layer to a feature space with unified dimensions, yielding the aligned feature $\hat{h}_i = W_i(h_i)$.
2) *Projection Head P_i:* A linear layer that projects $\hat{h}_i$ into a low-dimensional embedding space, obtaining the embedding vector $p_i = P_i(\hat{h}_i)$.
3) *Classification Head Y_i:* A linear classifier that takes $\hat{h}_i$ as input and computes the corresponding predicted class probabilities $y_{i,c} = Y_i(\hat{h}_i)$.

In each incremental stage, for all training samples (including new task data and memory exemplars) fed into the model, we perform dual-branch optimization on the output of each adapter. We apply the standard cross-entropy loss $\mathcal{L}_{CE}$ to the outputs of the classification heads of all adapters to supervise the model in learning all currently visible classes

$$\mathcal{L}_{CE} = -\sum_{i=1}^{l} \sum_{c=1}^{C_t} y_c \log \hat{y}_{i,c} \tag{4}$$

where l is the number of network layers, C_t is the number of currently visible classes, $y_c \in \{0, 1\}$ denotes the one-hot ground-truth label for class c, and $\hat{y}_{i,c}$ is the predicted probability for class c produced by the classification head of the ith layer.

In the projection space, SCL is applied to embedding vectors from the same batch to enhance intraclass compactness and interclass separability. Given a batch of samples with size B, we generate a copy via data augmentation (e.g., random cropping and rotation), forming an effective batch of size $2B$.

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For any sample x_k in the batch, the loss for all adapters is defined as

$$\mathcal{L}_{SCL} = -\frac{1}{2B}\frac{1}{|G_k|}\sum_{i=1}^{l}\sum_{k=1}^{2B}\sum_{g\in G_k}\log\frac{\exp\left(p_{i,k}\cdot p_g/\tau\right)}{\sum_{j\notin G_k}\exp\left(p_{i,k}\cdot p_j/\tau\right)} \quad (5)$$

where $p_{i,k}$ represents the projected representation of sample x_k, G_k denotes the set of samples in the batch that belong to the same class as x_k (including the original and augmented samples), p_g is the projection of a positive sample, and p_j is the projection of any other sample in the batch. The parameter τ is a temperature coefficient controlling the smoothness of the feature distribution.

2) IntraHD Module: The IntraHD module integrates two complementary mechanisms: FRD and CRD, as illustrated in Fig. 1.

FRD aims to provide reverse guidance to shallow features using deep features, forcing the shallow network to mimic the feature distribution of the deep network. Specifically, we align the feature $\hat{h}_l$ output by the deep network (via alignment layer W_l) with the feature $\hat{h}_i$ output by the shallow network (via alignment layer $W_i, i \in \{1,\ldots,l-1\}$)

$$\mathcal{L}_{FRD} = \frac{1}{2B}\sum_{i=1}^{l-1}\left\|\hat{h}'_l - \hat{h}'_i\right\|^2 \quad (6)$$

where $\hat{h}'_l$ and $\hat{h}'_i$ denote the normalized results of h'_l and h'_i, respectively.

CRD focuses on utilizing the output probability of the deep classifier to guide the learning of shallow classifiers, ensuring consistency in classification logic. We use the probability vector y_l output by the deep classification head Y_l as a soft label to constrain the output distribution y_i of shallow classification heads Y_i

$$\mathcal{L}_{CRD} = \frac{1}{2B}\sum_{i=1}^{l-1}\mathrm{KL}\left(y_i\|y_l\right) \quad (7)$$

where $y_i = \mathrm{softmax}(Z_i/T_{kd})$ and $y_l = \mathrm{softmax}(Z_l/T_{kd})$ are the temperature-scaled soft predictions of the ith shallow classifier and the deep classifier, respectively; Z_i denotes the logits of the ith classification head; and T_{kd} is the distillation temperature. The total loss of the IntraHD module is defined as

$$\mathcal{L}_{IntraHD} = \alpha\mathcal{L}_{FRD} + (1-\alpha)\mathcal{L}_{CRD} + \mathcal{L}_{CE} + \mathcal{L}_{SCL} \quad (8)$$

where α is a weighting coefficient balancing feature- and classification-level reverse distillation. Through the IntraHD module, the deep network provides comprehensive and precise supervision signals to the shallow network, significantly improving knowledge transfer efficiency and feature consistency.

3) InterKD Module: The InterKD module is the core component for preserving historical knowledge in the TGDD framework, as shown in Fig. 1.

To achieve effective cross-task knowledge transfer during the incremental stage, we design a knowledge transfer unit U_i, which adopts the same structure and dimension settings as the historical task adapters. This component aims to map the features extracted by the new task model F_θ to the same dimensions as the features extracted by the corresponding layers of the old task model $\tilde{F}_\theta$. This design strategy effectively decouples new task learning from historical knowledge retention. It ensures that the new task adapters focus solely on learning the new task, avoiding interference with historical knowledge migration, while the knowledge transfer unit is responsible for effectively transferring historical task knowledge to the new model.

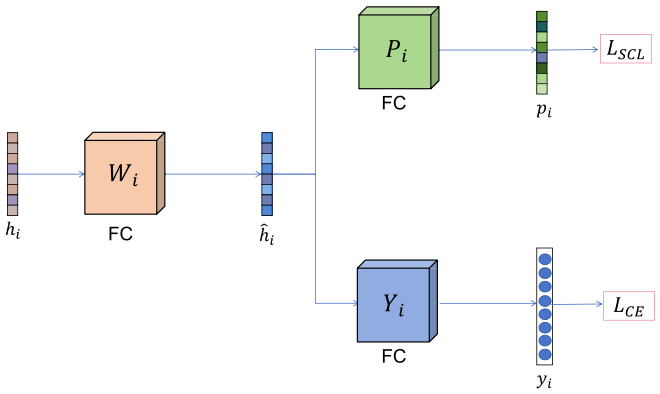


Fig. 2. Structure of the adapter module M_i, which maps the ith backbone feature h_i through a dimension alignment layer W_i to the unified dimension feature $\hat{h}_i$ and then feeds it to a projection head P_i and a classification head Y_i to produce the embedding p_i and the class probabilities y_i, respectively. It serves as the bridge connecting the features of different backbone layers to the downstream tasks and provides the shared building block on which both the intertask and intratask distillation are built.

To enforce effective transmission of historical knowledge, we introduce the InterKD loss $\mathcal{L}_{InterKD}$, which aligns the current task features p_i with the historical task features q_i

$$\mathcal{L}_{InterKD} = \frac{1}{M}\sum_{i=1}^{l}\left\|p'_i - q'_i\right\|^2 \quad (9)$$

where p'_i and q'_i are the normalized results of the p_i and q_i, respectively. By minimizing the Euclidean distance between them, $\mathcal{L}_{InterKD}$ ensures that the model retains historical knowledge while learning new tasks, effectively preventing catastrophic forgetting.

C. MSG Module

Hyperspectral incremental classification faces a unique challenge: classes introduced at different incremental stages are often confused due to high spectral similarity (e.g., "Forest" and "Shrubs"), yet they exhibit essential differences at the semantic level. If constraints are imposed solely within the visual feature space, the model struggles to establish clear decision boundaries when the spectra of new and old classes are highly similar; the representations of new classes can easily overwrite or interfere with the features of old classes, causing category boundaries to blur progressively with increments. To this end, we design the MSG module, as shown in Fig. 3. Using the text embeddings of class names as strong semantic priors, MSG actively injects semantic discriminative information into the visual feature space through explicit text–image feature alignment, ensuring that spectrally similar

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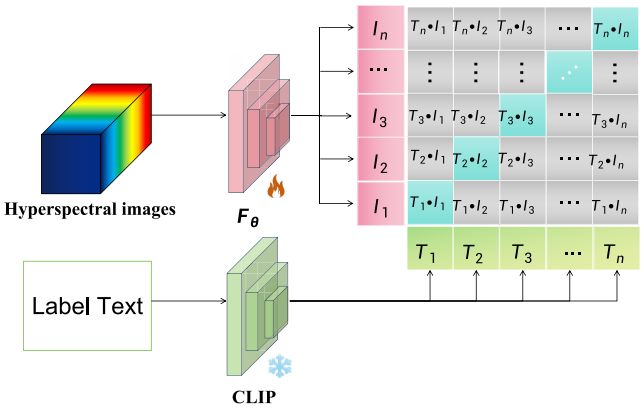


Fig. 3. Structure of the MSG module, which encodes HSI patches and class-name label texts via the trainable image encoder F_θ and the frozen CLIP text encoder and aligns the matched image–text pairs through a contrastive loss to separate spectrally similar but semantically distinct classes.

but semantically distinct classes maintain a sufficient discriminative distance in the representation space, thereby alleviating interclass confusion and catastrophic forgetting.

The core innovation of MSG lies in synergistically modeling image spectral features and the text embeddings of class names, leveraging this cross-modal semantic separability to guide the model in learning essential distinctions between classes.

Specifically, we construct a feature matching matrix to quantify cross-modal similarity, where the element $I_i \cdot T_j$ represents the correlation strength between image feature I_i and text feature T_j. During training, the model optimizes an alignment loss function to maximize the similarity of same-class image–text pairs while suppressing the similarity of different-class pairs. The loss function adopts a contrastive learning paradigm

$$\mathcal{L}_{MSG} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp\left(\text{sim}\left(I_i, T_i\right)/\gamma\right)}{\sum_{j=1}^{N}\exp\left(\text{sim}\left(I_i, T_j\right)/\gamma\right)} \tag{10}$$

where $\text{sim}(\cdot,\cdot)$ denotes the cosine similarity function, γ is a temperature hyperparameter, and N is the batch size.

D. Optimization Objective and Algorithm Workflow

Integrating the aforementioned modules, the TGDD framework utilizes an end-to-end training manner to synergistically optimize intertask knowledge transfer, intratask feature consistency, and cross-modal semantic alignment. To balance the retention of old knowledge while learning new tasks, we construct a joint optimization objective function. The total loss function $\mathcal{L}_{total}$ is composed of the InterKD loss $\mathcal{L}_{InterKD}$, the IntraHD loss $\mathcal{L}_{IntraHD}$, and the MSG loss $\mathcal{L}_{MSG}$, defined as follows:

$$\mathcal{L}_{total} = \mathcal{L}_{InterKD} + \mathcal{L}_{IntraHD} + \mathcal{L}_{MSG}. \tag{11}$$

It is worth noting that in the base task phase, since no historical tasks exist, $\mathcal{L}_{InterKD}$ is not involved in the calculation. The entire incremental learning process is formalized in Algorithm 1. The training process is divided into a base phase and an incremental phase. In the base phase, the model utilizes only IntraHD and MSG for initial learning and establishes an initial exemplar set (memory pool). In each subsequent incremental stage, the model receives new task data and replay samples from the memory pool, dynamically updates parameters by minimizing $\mathcal{L}_{total}$, and continuously updates the exemplar set using the sample selection strategy described in Section III-A to adapt to the evolving data stream.

Algorithm 1 Training Process of TGDD

Stage 1: Base Task Phase
Input:
The image feature extractor F_0
The text feature extractor CLIP
The initial memory data set $\mathcal{M}^0 = \emptyset$
The base task data label text set $\mathcal{T}_1$
Output:
The loss of model
for $x \in \mathcal{D}_{base}$ **do**
  Calculate the $\ell_{IntraHD}$ by Eq. (8)
  Calculate the ℓ_{MSG} by Eq. (10)
  $\ell_{total} = \ell_{IntraHD} + \ell_{MSG}$
  Update F_0 using ℓ_{total}
  Update $\mathcal{M}^0$ by Eq. (3)
end for
Stage 2: Incremental Task Phase
Input:
The memory data set $\mathcal{M}^1$
The image feature extractor F_1
The text feature extractor CLIP
The number of incremental phase T
Output:
The loss of model
for t in {2,..., T} **do**
  Obtain new data $\mathcal{D}_t$
  Obtain Task data label text $\mathcal{T}_t$
  for $x \in \mathcal{D}_t$ **do**
    for $\hat{x} \in \mathcal{M}_{t-1}$ **do**
      Calculate the $\ell_{InterKD}$ by Eq. (9)
    end for
      Calculate the $\ell_{IntraHD}$ by Eq. (8)
      Calculate the ℓ_{MSG} by Eq. (10)
      $\ell_{total} = \ell_{InterKD} + \ell_{IntraHD} + \ell_{MSG}$
      Update F_{t-1} using ℓ_{total}
      Update $\mathcal{M}^{t-1}$ by Eq. (3)
  end for
end for

IV. Experimental Results and Analysis

A. Datasets

To evaluate the proposed method, four benchmark HSI datasets covering different geographical regions and land cover types were adopted.

1) Pavia University: This dataset was acquired by the ROSIS sensor over an urban area in Pavia, northern Italy. The sensor covers a spectral range of 430–860 nm. After removing noisy channels, 103 spectral bands were retained. The image contains 610×340 pixels with a spatial resolution

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TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES

Class	PaviaU		Indian Pines		Salinas		Houston13	
	Train	Test	Train	Test	Train	Test	Train	Test
0	66	1327	1	10	20	402	12	251
1	186	3730	15	286	37	746	12	251
2	20	420	9	166	19	396	6	140
3	30	613	3	48	13	279	12	249
4	13	269	5	97	26	536	12	249
5	50	1006	8	146	39	792	3	65
6	13	266	1	6	35	716	12	254
7	36	737	5	96	112	2255	12	249
8	9	190	1	4	62	1241	12	251
9	-	-	10	195	32	656	12	246
10	-	-	25	491	10	214	12	247
11	-	-	6	119	19	386	12	247
12	-	-	3	41	9	184	4	94
13	-	-	13	253	10	214	4	86
14	-	-	4	78	72	1454	6	132
15	-	-	1	19	18	362	-	-

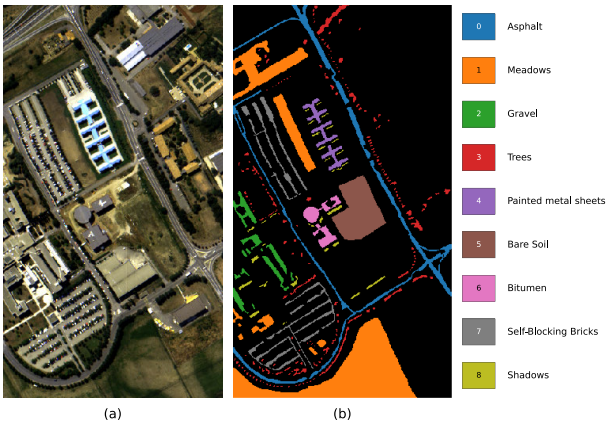


Fig. 4. (a) Pseudo-color image and (b) ground-truth map of PU.

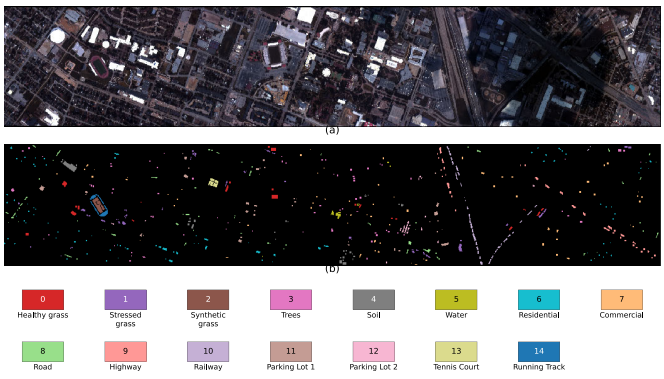


Fig. 5. (a) Pseudo-color image and (b) ground-truth map of HU13.

of 1.3 m/pixel. The ground truth includes nine different land cover classes, mainly characterized by urban materials and vegetation types. The class names and corresponding sample distributions are detailed in Table I. The pseudo-color image and ground-truth map are shown in Fig. 4.

2) Houston 2013 (HU13): This dataset was captured by the ITRES CASI (CASI-1500) sensor over the University of Houston campus and its surrounding urban areas in Texas, USA. The image consists of 349 × 1905 pixels and possesses 144 bands spanning the visible to near-infrared spectrum. The ground reference data contains 15 land cover classes, representing complex urban features including various vegetation types, construction materials, and infrastructure. A summary of sample statistics for each class is provided in Table I. The pseudo-color image and ground-truth map are shown in Fig. 5.

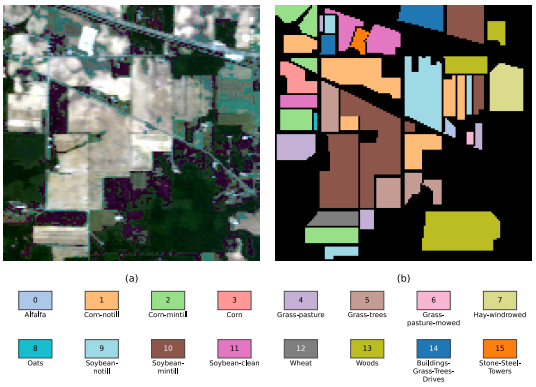


Fig. 6. (a) Pseudo-color image and (b) ground-truth map of IP.

3) Indian Pines: This dataset represents a mixed agricultural and forest landscape in Northwestern Indiana, USA, acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in June 1992. The scene consists of 145 × 145 pixels with a spatial resolution of 20 m/pixel. Out of the original 224 spectral bands covering the visible-to-mid-infrared wavelengths (400–2500 nm), 200 bands were retained after removing water absorption and low signal-to-noise ratio channels. The ground truth depicts 16 different land cover classes, primarily crops at different phenological stages and natural vegetation. Sample statistics are summarized in Table I. The pseudo-color image and ground-truth map are shown in Fig. 6.

4) Salinas: This dataset was collected by the AVIRIS sensor in Salinas (SA) Valley, California, USA. The image size is

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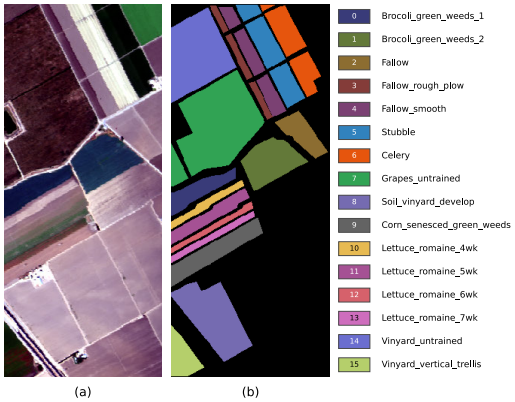


Fig. 7. (a) Pseudo-color image and (b) ground-truth map of SA.

512×217 pixels with a spatial resolution of 3.7 m/pixel. From the original 224 spectral bands, 204 bands were used after discarding water absorption bands (108–112, 154–167, and 224). The ground truth identifies 16 crop classes, representing various vegetables and vineyards at different maturity stages. The number of samples for each class is listed in Table I. The pseudo-color image and ground-truth map are shown in Fig. 7.

B. Experimental Setups

1) Data Preprocessing: In the experimental setup, the four HSI datasets were organized into a sequential continual learning task order. A partition strategy using 1% of samples for training and 20% for testing was adopted. Addressing the heterogeneity in spectral dimensions across datasets (ranging from 103 to 204 bands), this study first utilized principal component analysis (PCA) to reduce all datasets to 15 principal components. This ensures consistency in input dimensions and reduces computational overhead. Subsequently, to fully mine the joint spatial–spectral features of HSIs, a sliding window of 9×9 pixels was used to extract image patches centered on target pixels. These spectral–spatial patches served as network inputs, with the center pixel's label as the supervisory signal. This approach effectively fuses local spatial context while preserving spectral information, thereby enhancing feature discriminability and robustness.

2) Incremental Learning Settings: We designed a CIL scenario containing the four datasets. The number of classes for each dataset is 9, 15, 16, and 16, totaling 56 classes. The incremental learning process proceeds sequentially by dataset, with each stage introducing all classes of a new dataset. To ensure reproducibility and fairness, we followed the experimental settings of CSSFD. Specifically, 1% of labeled samples were randomly selected from each dataset for training, and 20% were used for testing. The remaining samples were not involved in model training or evaluation.

Regarding the DAES strategy, we set the total memory capacity to $M = 200$. For each learned class, the number of storable samples is calculated as $\lfloor M/n_c \rfloor$, where n_c is the current cumulative number of classes. If the number of available samples for a class is less than this value, all samples of that class are retained; otherwise, the proposed DAES strategy is used to select the most representative samples. This dynamic allocation strategy ensures a balanced representation of samples from each class in the memory buffer.



Fig. 8. Combined effect of joint supervised contrastive temperature, distillation temperature, and image–text contrastive in TGDD determines the resulting average OA heatmap.

TABLE II
PARAMETER TUNING RESULTS FOR α IN THE TGDD METHOD ACROSS THE FOUR DATASETS

Step	α					
	0	0.1	0.2	0.3	0.4	0.5
1	0.9811	0.9810	0.9812	0.9802	**0.9813**	0.9799
2	0.9444	0.9448	0.9436	0.9436	**0.9454**	0.9440
3	0.9552	0.9566	0.9549	0.9568	**0.9585**	0.9560
4	0.9318	0.9343	0.9345	0.9356	**0.9380**	0.9359
Total	3.8125	3.8167	3.8142	3.8162	**3.8232**	3.8158

3) Evaluation Metrics: To comprehensively evaluate model performance, we employed overall accuracy (OA), average accuracy (AA), and Kappa coefficient (KC) as basic evaluation metrics. Given the specific nature of incremental learning—where the model must maintain memory of past knowledge while continuously acquiring new knowledge—we further introduced average forgetting (AF) as a key supplementary metric. AF quantifies the extent of forgetting of previous task knowledge during the learning of new tasks. This provides a multidimensional and systematic measurement framework for assessing the comprehensive performance of incremental learning models. The formula for AF is

$$AF = \frac{1}{T-1} \sum_{i=1}^{T-1} \left(\max_{j \in \{i,\ldots,T-1\}} A_{i,j} - A_{i,T} \right) \tag{12}$$

where T is the total number of tasks in incremental learning and $A_{i,j}$ represents the classification accuracy of the model on the test set of the ith task after learning the jth task. AF is a crucial indicator for assessing the performance of incremental learning algorithms.

4) Parameter Configuration: The experiments were implemented based on the PyTorch framework. The Adam optimizer

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TABLE III
PERFORMANCE COMPARISON BETWEEN TGDD AND ALTERNATIVE SAMPLING METHODS

	RS	US	PS	HS	TGDD
Step1	**98.13**	**98.13**	**98.13**	**98.13**	**98.13**
Step2	94.27	92.46	93.95	94.21	**94.54**
Step3	94.63	92.90	94.48	94.63	**95.85**
Step4	92.06	89.90	91.96	91.87	**93.80**

was used for model training, with an initial learning rate of 0.01, a weight decay coefficient of 0.001, 100 training epochs, and a batch size of 64. For key hyperparameters in the loss function, a grid search strategy was employed for fine-tuning. The supervised contrastive temperature τ was searched in $\{0.1, 0.3, 0.5, 0.7, 0.9\}$; the distillation temperature T_{kd} in $\{1.0, 2.0, 3.0, 4.0\}$; and the contrastive temperature γ in $\{0.05, 0.07, 0.10, 0.12, 0.15\}$. The optimal parameter combination was determined as $\tau = 0.7$, $T_{kd} = 2.0$, and $\gamma = 0.12$. As shown in Fig. 8, under this configuration, the model achieved an average classification accuracy of 95.58% across four datasets. For the weighting parameter α in the IntraHD module, we verified it in the interval $\{0.1, 0.2, 0.3, 0.4, 0.5\}$. As shown in Table II, the model achieved optimal comprehensive performance across four incremental stages when $\alpha = 0.4$. The accuracies for each stage were 98.13%, 94.54%, 95.85%, and 93.80%. This indicates that moderately increasing the weight of classification-level distillation effectively promotes knowledge transfer and retention. The text encoder was initialized using the CLIP pretrained ViT-B/32 model to fully leverage the semantic prior knowledge obtained from large-scale vision-language pretraining. All experiments were conducted on a high-performance workstation equipped with an Intel Xeon W7-3445 CPU (2.9 GHz), 256-GB RAM, and an NVIDIA RTX 6000 Ada GPU (48-GB VRAM).

C. Results and Analysis

1) Analysis of Sample Selection Strategies: This article adopts a DAES strategy, aiming to retain the most representative samples from each class. To better evaluate the effectiveness of this strategy, we compared it with several common sample selection methods.

1) *RS:* Randomly select a fixed number of samples from all old samples. This method is simple to implement but unstable in effect and difficult to ensure representativeness.
2) *Uncertainty-Based Sampling (US):* Retain samples where the model's prediction is most "uncertain" (e.g., samples with high entropy). While helpful for retaining knowledge the model easily forgets, it requires forward inference and has high computational complexity.
3) *Prototype Selection (PS):* Also known as Herding, it selects samples close to the class centroid in the feature space. This is a classic method used in iCaRL and preserves the main characteristics of a class well.
4) *Hardness Sampling (HS):* Select samples that are misclassified or have low confidence, focusing on boundary samples that are easily confused.

Based on the analysis of results in Table III, the proposed DAES strategy demonstrates good performance across all incremental learning stages. In Step 1, since no sample retention is involved, all methods achieve an accuracy of 98.13%. Starting from Step 2, where retention mechanisms are applied, TGDD achieves the highest accuracy of 94.54%, while US achieves the lowest at 92.46%. In Step 3, the advantage of DAES becomes more pronounced, with an accuracy of 95.85%, surpassing the second-best method by 1.22%. In Step 4, although accuracies for all methods decline, TGDD maintains the highest level at 93.80%, whereas US is the lowest at 89.90%.

Looking at the overall trend, the performance of all methods decreases as incremental steps increase, reflecting the inherent catastrophic forgetting in incremental learning. However, TGDD exhibits the smallest decline (only 4.33%), while US shows the largest decline (8.23%). This indicates that DAES effectively retains the most representative samples, thereby alleviating knowledge forgetting. RS, despite its simplicity, performs relatively stably. The poor performance of US suggests that focusing solely on uncertain samples is detrimental to overall performance retention. PS and HS perform similarly but are inferior to TGDD. In summary, the results fully prove that an effective sample selection strategy is crucial for maintaining model performance in incremental learning, and DAES possesses significant advantages in retaining representative samples and mitigating catastrophic forgetting. To provide more intuitive evidence at the feature-distribution level, beyond the accuracy-based comparison in Table III, we further compare the five sampling strategies using t-SNE visualizations together with a quantitative coverage metric. Let $D = \{x_1, x_2, \ldots, x_{|D|}\}$ denote all original samples of a given class and let $S = \{s_1, s_2, \ldots, s_{|S|}\}$ denote the corresponding selected exemplar set. To quantitatively measure how faithfully the exemplars preserve the original feature distribution, we adopt a normalized Chamfer-style coverage metric

$$\text{Coverage}(S, D) = \frac{1}{|D|} \sum_{i=1}^{|D|} \frac{\min_{1 \le j \le |S|} \left\| x_i - s_j \right\|_2^2}{\max_{1 \le p,q \le |D|} \left\| x_p - x_q \right\|_2^2} \tag{13}$$

where the numerator measures the squared distance from each original sample x_i to its nearest selected exemplar s_j and the denominator normalizes this distance by the squared diameter of D. A smaller coverage value therefore indicates that the exemplars more faithfully cover the original feature structure. We choose two representative classes, asphalt and bare soil, for visualization. For each class, we plot the original t-SNE distribution of all samples together with the samples selected by RS, US, PS, HS, and DAES, highlighted in orange over the light-gray original distribution, and outline the convex hull of the selected samples with an orange dashed line for an intuitive view of their coverage.

As shown in Fig. 9, on both asphalt and bare soil, the exemplars selected by DAES are visually closest to the shape of the original t-SNE distribution. The orange convex hull of DAES approximately reproduces the multicluster substructure of the class, whereas RS, US, PS, and HS tend to collapse onto a single region or a narrow boundary band, leaving large

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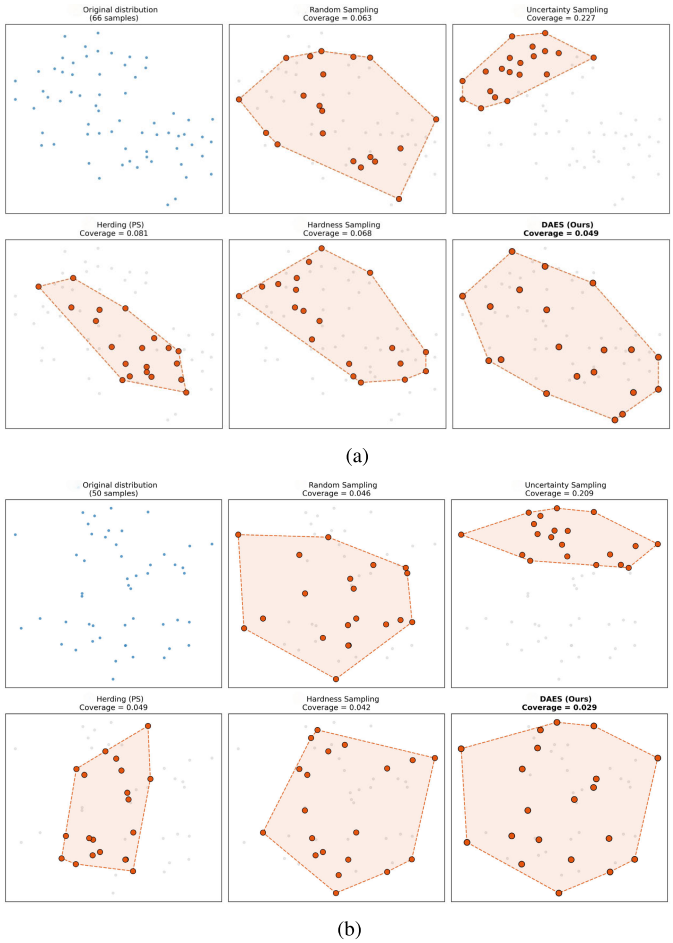


Fig. 9. t-SNE visualization of exemplar selection by the five sampling strategies on two representative classes. (a) Asphalt and (b) bare soil. For each class, from left to right, the six panels show the original t-SNE distribution of all samples in the class. The corresponding coverage values (lower is better) are annotated in each panel.

portions of the original distribution uncovered. Quantitatively, the coverage value of DAES is the lowest among the five strategies on both classes, which is in agreement with the visual observations. These results directly validate, at the feature-distribution level, that the multi-substructure-aware design of DAES is more effective for exemplar selection in hyperspectral incremental learning than strategies that implicitly assume a single compact distribution.

2) Comparison With State-of-the-Art Methods: To verify the classification performance of the proposed model, we compared it with six mainstream incremental learning methods—LwF [38], iCaRL [14], MOSE [31], CILKD [26], CSSFD [23], and HyperKD [51]—on the Pavia University (PU), Indian Pine (IP), SA, and HU13 datasets. To ensure fairness, all comparative experiments adopted the same experimental settings.

Table IV presents the results of each method on the four datasets. Taking the PU dataset in Step 1 as an example, the proposed TGDD method achieved an OA of 98.13%, outperforming LwF, iCaRL, MOSE, CILKD, CSSFD, and HyperKD by 3.54%, 0.65%, 1.59%, 2.80%, 3.44%, and 1.96%, respectively.



Fig. 10. Classification results of different methods on the PaviaU dataset, trained with 1% labeled samples and a memory buffer M = 200 samples. (a) LWF. (b) iCaRL. (c) MOSE. (d) CILKD. (e) CSSFD. (f) HyperKD. (g) OUR.

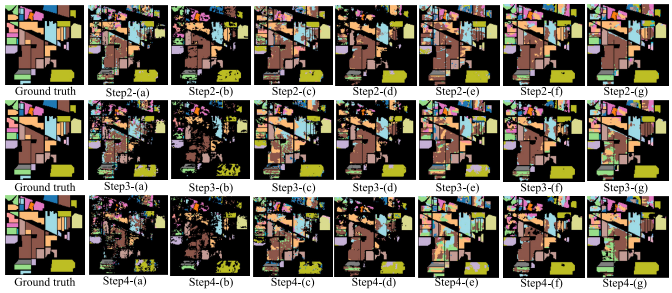


Fig. 11. Classification results of different methods on the IP dataset, trained with 1% labeled samples and a memory buffer M = 200 samples. (a) LWF. (b) iCaRL. (c) MOSE. (d) CILKD. (e) CSSFD. (f) HyperKD. (g) OUR.

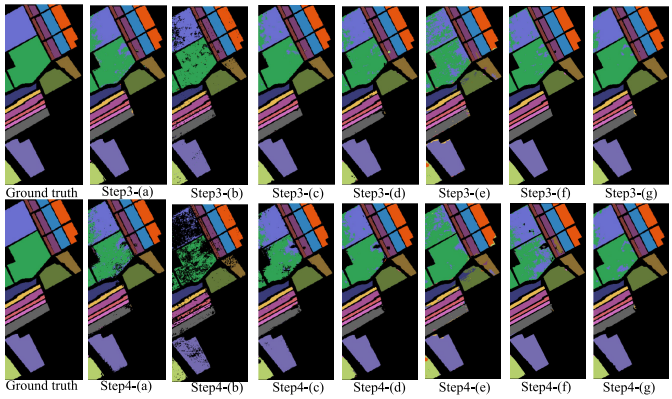


Fig. 12. Classification results of different methods on the SAs dataset, trained with 1% labeled samples and a memory buffer M = 200 samples. (a) LWF. (b) iCaRL. (c) MOSE. (d) CILKD. (e) CSSFD. (f) HyperKD. (g) OUR.

As incremental learning proceeded to Step 2 with the introduction of the IP dataset, TGDD achieved an OA of 79.98%. This represents a significant improvement over the other six methods, exceeding iCaRL, LwF, MOSE, CILKD, CSSFD, and HyperKD by 37.67%, 23.84%, 18.02%, 20.38%,

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TABLE IV
COMPARATIVE EXPERIMENTAL RESULTS (%) OF THE TGDD METHOD WITH OTHER METHODS ON FOUR DATASETS

Step	Data		Method						
	Dataset	Metric	LWF	iCaRL	MOSE	CILKD	CSSFD	HyperKD	TGDD
1	PU	OA	94.59±0.8	97.48±0.3	96.54±0.2	95.33±0.1	94.69±3.0	96.17±0.85	**98.13±0.1**
		AA	90.97±1.2	96.29±0.5	92.80±0.6	91.66±0.2	93.51±1.0	93.06±1.71	**96.65±0.1**
		KC	0.9	0.96	0.95	0.94	0.92	0.95	**0.97**
2	PU	OA	88.62±2.3	86.12±2.4	93.59±0.3	91.40±0.3	90.70±0.1	94.98±1.39	**98.03±0.1**
		AA	88.11±1.0	89.50±1.3	91.12±0.3	87.77±0.6	90.47±0.2	92.87±1.80	**96.69±0.2**
		KC	0.86	0.83	0.92	0.88	0.89	0.93	**0.97**
2	IP	OA	56.14±1.3	42.31±1.0	61.96±0.9	59.60±0.4	67.30±0.6	70.33±2.26	**79.98±0.5**
		AA	49.59±1.7	44.39±1.7	53.17±1.5	43.94±0.3	44.39±1.1	64.84±4.48	**80.47±0.7**
		KC	0.50	0.37	0.57	0.55	0.70	0.66	**0.77**
3	PU	OA	67.75±3.7	79.23±1.9	81.44±2.3	91.52±0.4	88.33±0.3	93.37±1.41	**96.90±0.2**
		AA	73.90±1.5	84.43±1.6	85.20±1.2	88.97±0.6	83.12±0.5	92.44±1.47	**95.63±0.2**
		KC	0.62	0.75	0.77	0.89	0.84	0.91	**0.96**
3	IP	OA	44.79±1.9	38.38±1.3	55.36±1.3	58.31±0.3	63.99±0.3	68.34±1.79	**78.92±0.7**
		AA	41.12±1.5	39.85±2.1	50.37±1.2	43.43±0.5	63.64±0.1	63.24±4.26	**81.43±0.9**
		KC	0.39	0.33	0.50	0.53	0.59	0.64	**0.76**
3	Sa	OA	92.83±0.5	85.73±0.5	95.67±0.3	95.18±0.1	88.42±0.4	96.31±0.27	**98.24±0.1**
		AA	94.88±0.7	86.96±0.5	97.54±0.3	96.48±0.2	91.41±0.3	98.07±0.16	**99.03±0.1**
		KC	0.92	0.84	0.95	0.94	0.87	0.96	**0.98**
4	PU	OA	53.42±2.6	68.92±3.5	68.68±1.6	89.53±0.3	87.30±1.2	79.04±7.10	**94.34±0.5**
		AA	64.27±2.1	79.08±1.6	76.50±1.1	87.78±0.3	82.14±0.6	83.91±3.65	**94.71±0.3**
		KC	0.47	0.64	0.62	0.86	0.83	0.74	**0.93**
4	IP	OA	36.01±1.4	34.04±1.3	48.41±1.1	54.16±0.7	60.57±0.8	62.07±2.50	**78.70±0.6**
		AA	34.28±1.5	37.08±2.3	48.33±1.2	40.49±0.7	60.03±0.9	55.74±2.88	**82.21±0.9**
		KC	0.31	0.29	0.43	0.49	0.55	0.58	**0.76**
4	Sa	OA	96.82±0.9	81.84±5.7	89.00±0.7	93.47±0.2	88.62±0.5	92.77±1.36	**97.36±0.1**
		AA	92.00±0.7	85.46±4.9	93.99±0.4	96.11±0.1	91.66±0.3	96.39±0.82	**98.64±0.1**
		KC	0.86	0.80	0.88	0.92	0.87	0.92	**0.97**
4	HU13	OA	79.19±0.6	64.43±1.6	83.64±0.3	67.01±1.8	70.03±0.3	88.08±1.38	**89.77±0.5**
		AA	75.98±0.6	65.74±1.5	83.64±0.5	65.79±1.8	65.33±1.3	88.79±0.94	**90.78±0.4**
		KC	0.78	0.60	0.82	0.65	0.68	0.87	**0.89**

12.68%, and 9.65%, respectively. In terms of AA, TGDD reached 80.47%, still 15.63% higher than the best-performing competitor, HyperKD. Meanwhile, TGDD excelled in preserving old task performance. From Step 1 to Step 2, TGDD's performance on the PU dataset remained relatively stable with an OA of 98.03% (a specific drop of only 0.1%), whereas other methods like LwF, iCaRL, MOSE, and HyperKD dropped by approximately 5.97%, 11.36%, 2.95%, and 1.19%, respectively. This demonstrates that TGDD effectively retains old knowledge while learning new tasks.

In Step 3, the SA dataset was introduced. TGDD showed strong adaptability, achieving an OA of 98.24%. Compared to LwF, iCaRL, MOSE, and HyperKD, it improved by 5.41%, 12.51%, 2.57%, and 1.93%, respectively. For AA, TGDD reached 99.03%, significantly outperforming iCaRL and CSSFD by 12.07% and 7.62%. The KC remained high at 0.98. For previously learned datasets, TGDD continued to perform excellently. On the PU dataset, it improved by 29.15%, 17.67%, 15.46%, 5.38%, 8.57%, and 3.53% compared to the six methods. On the IP dataset, the performance gain was distinct, exceeding the benchmarks by 34.13%, 40.54%, 23.56%, 20.61%, 14.93%, and 10.58%.

Finally, in Step 4, the HU13 dataset (15 classes) was introduced, requiring the model to classify a total of 56 classes. TGDD achieved an OA of 89.77% on the new task (HU13), outperforming LwF, iCaRL, MOSE, CILKD, CSSFD, and HyperKD by 10.58%, 25.34%, 6.13%, 22.76%, 19.74%, and 1.69%, respectively. Comparing previous tasks, TGDD maintained its lead. On PU, it exceeded the competitors by 40.92%, 25.42%, 25.66%, 4.81%, 7.04%, and 15.30%. On IP, the improvement was even more significant (42.69%, 44.66%, 30.29%, 24.54%, 18.13%, and 16.63%). On SA, OA also improved by 0.54%–15.52%. These data fully illustrate that TGDD effectively maintains recognition capabilities for previous tasks after accumulating multiple tasks, demonstrating strong robustness and scalability.

3) Visualization Analysis: Figs. 10–13 display the visualization results of all methods across the four task steps. To clearly show the effects for each dataset, pixels predicted as classes from other datasets were set to black. Fig. 10 shows the results on the PU dataset. In the first basic task, all methods obtained coherent land cover results, though some misclassifications remained. LwF, iCaRL, MOSE, CILKD, CSSFD, and HyperKD showed more misclassifications compared to TGDD. This may be because TGDD uses a 3-D CNN to simultaneously capture spatial and spectral information, effectively fusing them to enhance the performance. In the second task (adding IP), noise in the classification maps increased, especially in the central building area. This

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TABLE V
ABLATION COMPARISON RESULTS (%) OF EACH VARIANT OF TGDD

#	Base	InterKD	IntraHD	MSG	Step1			Step2			Step3			Step4		
					OA	AA	KC	OA	AA	KC	OA	AA	KC	OA	AA	KC
1	✓				97.37	95.30	0.96	92.26	82.76	0.90	92.56	87.63	0.92	87.75	87.04	0.87
2	✓	✓			98.06	96.47	0.97	93.46	85.71	0.92	94.47	90.34	0.94	91.47	89.67	0.91
3	✓		✓		97.79	96.25	0.97	93.68	83.47	0.92	94.87	89.67	0.94	92.73	89.64	0.92
4	✓			✓	97.88	96.09	0.97	93.34	85.72	0.92	95.43	90.46	0.95	92.69	89.60	0.92
5	✓	✓	✓		98.11	96.57	0.97	93.98	85.81	0.92	95.13	91.36	0.94	92.35	90.64	0.92
6	✓	✓	✓	✓	**98.13**	**96.65**	**0.98**	**94.54**	**86.31**	**0.93**	**95.85**	**91.41**	**0.95**	**93.80**	**92.21**	**0.93**

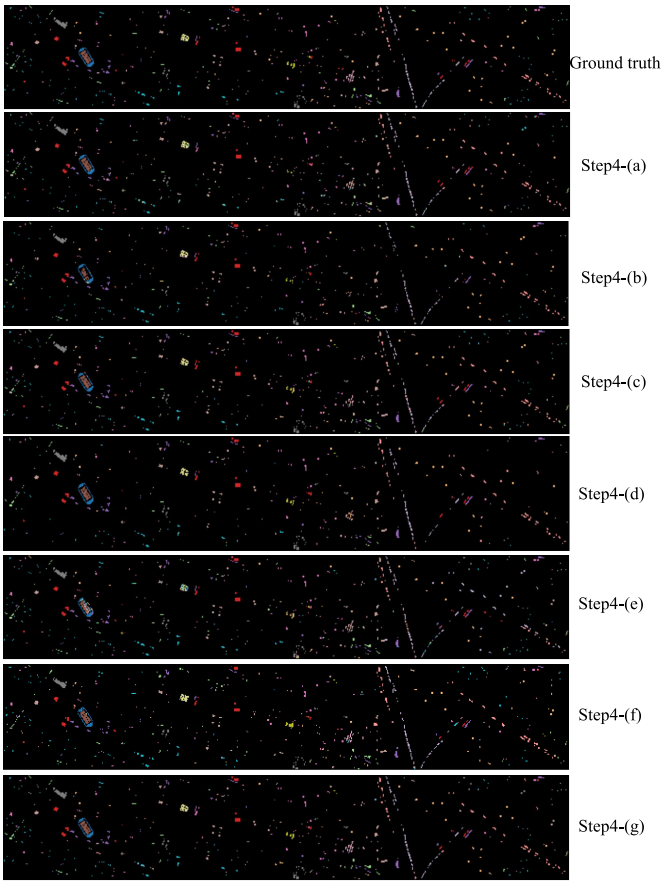


Fig. 13. Classification results of different methods on the Houston dataset, trained with 1% labeled samples and a memory buffer M = 200 samples. (a) LWF. (b) iCaRL. (c) MOSE. (d) CILKD. (e) CSSFD. (f) HyperKD. (g) OUR.

noise persisted in the third task. In contrast, TGDD produced classification maps with more regular shapes and less noise, indicating better classification effectiveness and robustness.

Furthermore, despite retaining historical knowledge, acquiring new knowledge rapidly from new datasets remains challenging. As shown in Figs. 11 and 12 (Steps 2 and 3), all methods exhibited numerous misclassifications upon adding the IP and SA datasets, suggesting insufficient learning of new tasks. Notably, LwF, iCaRL, and MOSE tended to misclassify samples from the new dataset as classes from other datasets, while CILKD and CSSFD tended to misclassify them as other classes within the same dataset.

In Fig. 13 (Step 4, HU13 dataset), many misclassifications persisted. For instance, iCaRL and CSSFD showed obvious errors in the left rectangular building area. In summary, as the model updated, comparative methods failed to consistently maintain good performance on the IPs dataset. The proposed strategy of introducing text modality alignment effectively alleviates cross-domain confusion and improves adaptability to new tasks.

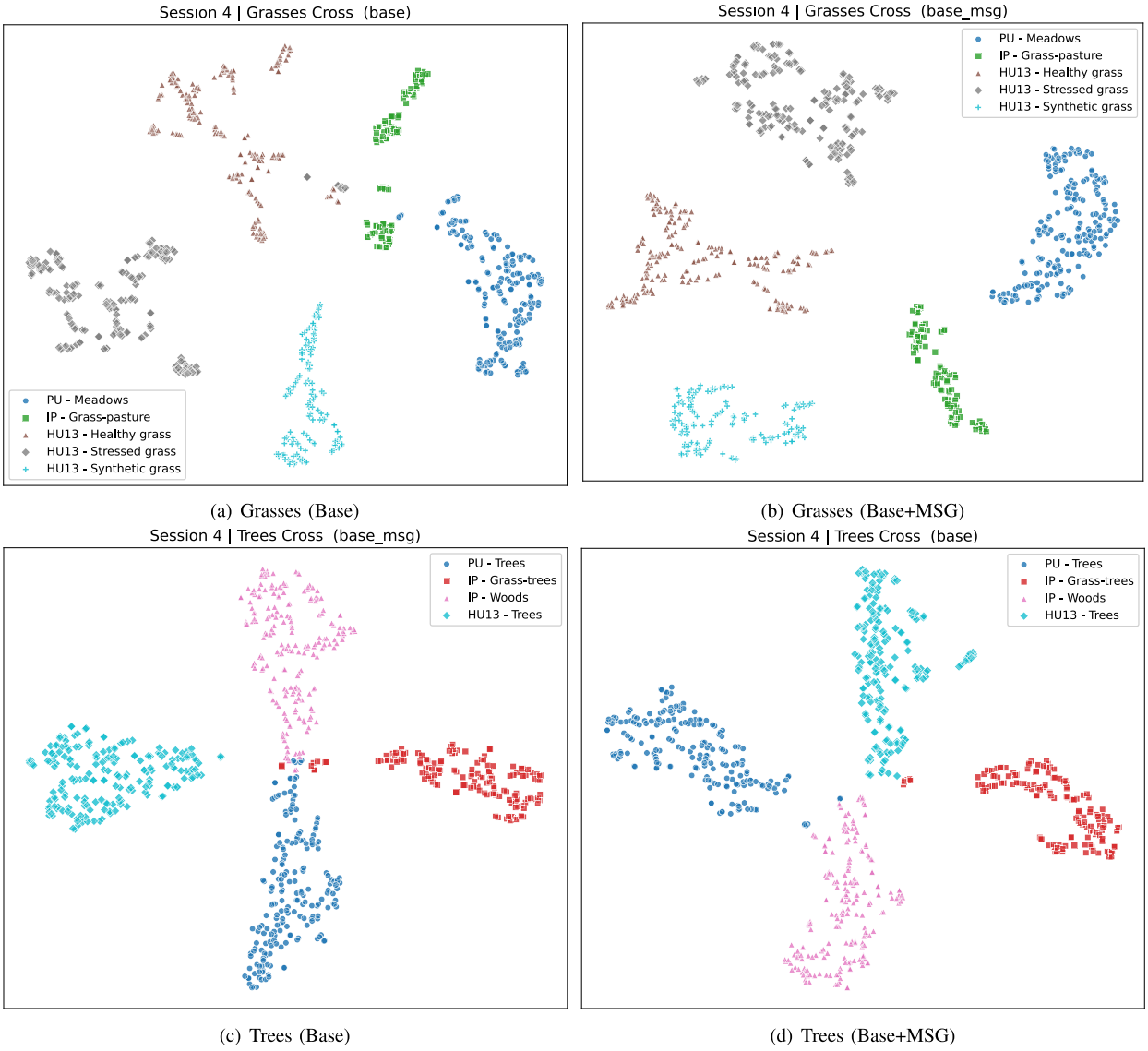
D. Ablation Study

To fully evaluate the effectiveness of the proposed method, we designed a series of ablation experiments with four incremental tasks on the four datasets. These experiments explored the impact of different loss modules: InterKD, IntraHD, and MSG, as shown in Table V.

The Baseline employs a 3-D CNN (six layers) trained with the DAES strategy. When using the Baseline alone, although Step 1 OA reached 97.37%, Step 4 OA dropped significantly to 87.75%, indicating severe catastrophic forgetting. Introducing the InterKD module significantly improved performance. Specifically, from Step 1 to Step 4, OA increased by 0.69%, 1.20%, 1.91%, and 3.72%, and AA increased by 1.17%, 2.95%, 2.71%, and 2.63%. This indicates that InterKD effectively transfers old task knowledge to the current task. Further adding the IntraHD module resulted in a continuous upward trend in performance, especially in later tasks. In Steps 3 and 4, OA increased by 0.66% and 0.88%, and AA increased by 1.02% and 0.97%. This proves that IntraHD's distillation mechanism from deep to shallow networks effectively enhances generalization. Finally, by integrating the MSG module, the complete TGDD framework achieved the best performance in all stages. Particularly in Step 4, OA further improved by 1.45%, and KC improved by 0.01. This strongly proves that MSG, by aligning text priors with image features, significantly enhances the discrimination of similar classes and mitigates the negative impact of class similarity on old task performance. Overall, as incremental steps deepen, the contribution of each module shows a clear cumulative effect. The complete model exhibits the most stable performance curve, validating the TGDD framework's effectiveness in balancing plasticity and stability.

To further isolate the individual contribution of each component, we additionally compared the variants that add only a single module to the Baseline (rows 2–4 in Table V). Each module independently brings consistent gains over the Baseline, and the improvement becomes more pronounced

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(a) Grasses (Base) (b) Grasses (Base+MSG)

(c) Trees (Base) (d) Trees (Base+MSG)

Fig. 14. t-SNE visualization of cross-dataset look-alike classes at Session 4, comparing the Base model with Base+MSG. (a) and (b) Five grass-related classes (PU-meadows, IP-grass-pasture, HU13-healthy grass, HU13-stressed grass, and HU13-synthetic grass). (c) and (d) Four tree-related classes (PU-trees, IP-grass-trees, IP-woods, and HU13-trees). Compared with the base model, Base+MSG produces clearly more compact intraclass clusters and substantially larger interclass margins on both groups.

as incremental steps accumulate. At Step 4, adding InterKD, IntraHD, or MSG alone raises the OA from 87.75% to 91.47%, 92.73%, and 92.69%, corresponding to gains of 3.72%, 4.98%, and 4.94%, respectively, with the corresponding AA also improving from 87.04% to 89.67%, 89.64%, and 89.60%. This demonstrates that all three modules are able to alleviate catastrophic forgetting on their own: InterKD transfers cross-task knowledge from the old model, while IntraHD and MSG, which operate within the current task, yield even larger long-term gains by enforcing hierarchically consistent features and injecting semantic discrimination, respectively. When the modules are combined, their effects prove complementary rather than redundant, and the full TGDD model consistently achieves the best results across all four steps (e.g., 93.80% OA at Step 4), confirming that the three components mutually reinforce one another.

To further verify the effectiveness of the MSG module in distinguishing spectrally similar classes across different datasets, we visualize the feature distributions of the Base and Base+MSG models with t-SNE on two representative groups of cross-dataset look-alike classes at Session 4: a grass group of five classes (PU-meadows, IP-grass-pasture, HU13-healthy grass, HU13-stressed grass, and HU13-synthetic grass) and a tree group of four classes (PU-trees, IP-grass-trees, IP-woods, and HU13-trees). As shown in Fig. 14, in the Base model, several classes from the same group heavily intertwine in the feature space. For the grass group, PU-Meadows, IP-Grass-pasture, and HU13-Healthy grass are largely mixed together; for the tree group, PU-Trees overlaps substantially with IP-Woods and IP-Grass-trees in the central region. After introducing MSG, all classes in both groups aggregate into clearly compact

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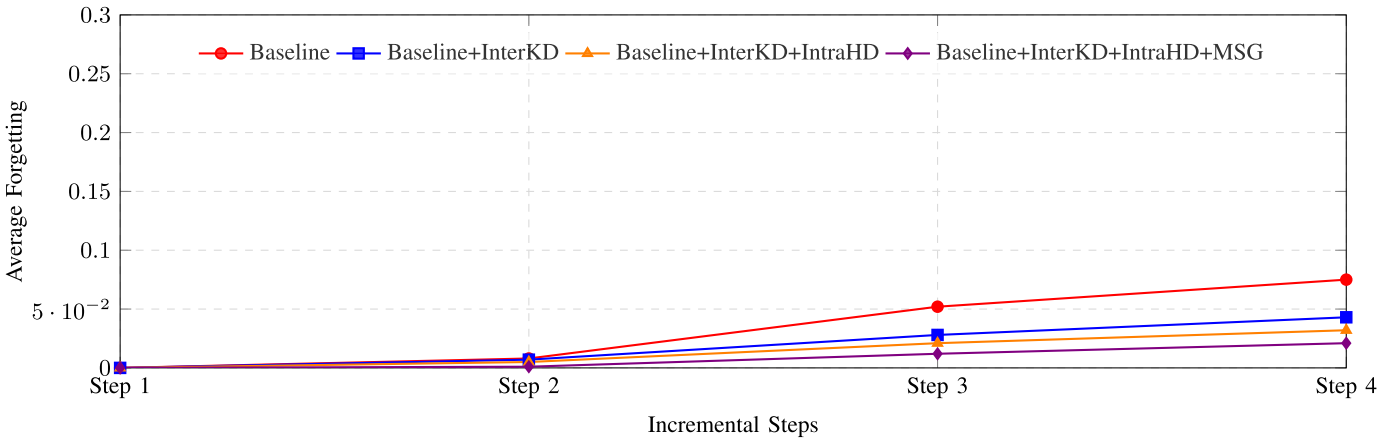


Fig. 15. Schematic of AF curve.

clusters with substantially enlarged intercluster margins. This intuitively confirms that MSG, by aligning visual features with the relative semantic distances captured by the CLIP text encoder, indeed provides a discriminative gain that cannot be obtained by relying solely on spectral features, thereby effectively mitigating cross-dataset class confusion in hyperspectral incremental learning.

To verify the impact of each module on AF across incremental steps, we plotted the AF curve in Fig. 15. It is clearly observable that as more modules are progressively introduced, the AF rate shows a significant downward trend. Specifically, the Baseline model exhibits the highest forgetting rate. With the sequential addition of InterKD, IntraHD, and MSG modules, AF continuously decreases. The complete framework (Baseline + InterKD + IntraHD + MSG) demonstrates the lowest forgetting rate in all incremental steps. This indicates that the introduction of each additional module effectively enhances the model's ability to retain old knowledge while learning new tasks, successfully alleviating catastrophic forgetting.

V. Conclusion

This article proposed a TGDD framework, providing a systematic solution to catastrophic forgetting in HSI classification. The framework effectively learns new tasks while maintaining the stability of old knowledge through the synergy of four core modules. First, the DAES strategy effectively maintains the sample diversity of old classes under limited storage. Compared to RS, it avoids class imbalance and lays a solid foundation for subsequent tasks. Second, the InterKD module successfully achieves efficient transfer of historical knowledge by aligning the feature spaces of the old and current models, significantly reducing the risk of forgetting. Third, the IntraHD module enables the model to learn highly discriminative and hierarchically consistent feature representations, improving generalization to new tasks. Finally, the MSG module dynamically adjusts feature learning and decision boundaries by introducing text embeddings of class names, effectively mitigating class confusion caused by spectral similarity. This ensures that the model accurately recognizes old classes while learning new ones. In summary, by integrating sample selection, KD, and multimodal learning, TGDD demonstrated superior classification accuracy and robustness in experiments on multiple public HSI datasets. This proves its immense potential for effective incremental learning in the field of hyperspectral remote sensing. Nevertheless, the current design still has two structural limitations. First, the text encoder relied upon by MSG is pretrained on general vision-language data, resulting in insufficient semantic discriminability for extremely fine-grained categories within the remote sensing domain. Second, the existing uniform storage allocation strategy fails to consider the significant differences among various classes in terms of forgetting rate, intraclass complexity, and interclass confusion, leading to a suboptimal allocation of storage resources. To address the former, future work will explore domain-adaptive prompt tuning and remote sensing-corpus-driven text encoder adaptation tailored for hyperspectral scenarios. To address the latter, we will investigate risk-aware adaptive sample selection mechanisms to dynamically allocate storage quotas based on forgetting risks and discriminative difficulties, thereby further improving memory efficiency. Furthermore, extending this framework to exemplar-free and open-set incremental scenarios is a crucial direction for enhancing its applicability in real-world deployments.

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